

1 Introduction

The present notes outline the evolution of a system with a constant magnitude.

Example 1 With any $\omega > 0$, the function $w : \mathbb{R} \rightarrow \mathbb{R}^2$ defined by

$$w(t) := \begin{bmatrix} u(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} \cos(\omega \cdot t) \\ \sin(\omega \cdot t) \end{bmatrix}$$

parametrizes a planar curve that stays at a constant distance, or magnitude, away from $\mathbf{0}$.

The functions u and v and hence also w are solutions of the differential equation

$$u'' = -\omega^2 \cdot u, \quad v'' = -\omega^2 \cdot v, \quad w'' = -\omega^2 \cdot w.$$

Complex numbers allow for a different differential equation: express $w : \mathbb{R} \rightarrow \mathbb{C}$ as

$$w(t) = u(t) + i \cdot v(t) = \cos(\omega \cdot t) + i \cdot \sin(\omega \cdot t).$$

The first derivatives is

$$w'(t) = i \cdot v'(t) + u'(t) = i \cdot \omega \cdot \cos(\omega \cdot t) - \omega \cdot \sin(\omega \cdot t) = i \cdot \omega \cdot w(t).$$

Thus, the function w is a solution of the linear first-order differential equation

$$-\frac{i}{\omega} \cdot w' = w.$$

What linear first-order differential equations have periodic solutions with constant magnitude?

Up to a multiplicative constant $\hbar > 0$, with a matrix $H \in \mathbb{M}_{2 \times 2}(\mathbb{C})$, such an equation may be

$$-i \cdot \hbar \cdot w' = H \cdot w,$$

where the matrix H preserves the norm on $\mathbb{C}^2$, so that $\|H \cdot w\| = \|w\|$ for every $w \in \mathbb{C}^2$. Equivalently, in terms of inner products, $\langle w, w \rangle = \|w\|^2 = \|H \cdot w\|^2 = \langle H \cdot w, H \cdot w \rangle = w^* \cdot H^* \cdot H \cdot w$.

2 Isometries

This section outlines some features of transformations that preserve magnitudes.

Theorem 2 (Polar Identity) *For every inner product and its induced norm,*

$$\langle \mathbf{z}, \mathbf{w} \rangle = \frac{\|\mathbf{z} + \mathbf{w}\|^2 - \|\mathbf{z} - \mathbf{w}\|^2}{4} - i \cdot \frac{\|\mathbf{z} + i \cdot \mathbf{w}\|^2 - \|\mathbf{z} - i \cdot \mathbf{w}\|^2}{4}$$

Proof. First, apply the definition of the induced norm, $\|\mathbf{t}\|^2 := \langle \mathbf{t}, \mathbf{t} \rangle$ and $\|\mathbf{t}\| \geq 0$, to $\mathbf{t} := \mathbf{z} \pm \mathbf{w}$:

$$\begin{aligned} \|\mathbf{z} + \mathbf{w}\|^2 &= \langle \mathbf{z} + \mathbf{w}, \mathbf{z} + \mathbf{w} \rangle \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle + \langle \mathbf{z}, \mathbf{w} \rangle + \langle \mathbf{w}, \mathbf{z} \rangle \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle + \langle \mathbf{z}, \mathbf{w} \rangle + \overline{\langle \mathbf{z}, \mathbf{w} \rangle} \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle + 2 \cdot \Re \langle \mathbf{z}, \mathbf{w} \rangle \\ \|\mathbf{z} - \mathbf{w}\|^2 &= \langle \mathbf{z} - \mathbf{w}, \mathbf{z} - \mathbf{w} \rangle \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle - \langle \mathbf{z}, \mathbf{w} \rangle - \langle \mathbf{w}, \mathbf{z} \rangle \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle - \langle \mathbf{z}, \mathbf{w} \rangle - \overline{\langle \mathbf{z}, \mathbf{w} \rangle} \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle - 2 \cdot \Re \langle \mathbf{z}, \mathbf{w} \rangle \\ \|\mathbf{z} + \mathbf{w}\|^2 - \|\mathbf{z} - \mathbf{w}\|^2 &= 4 \cdot \Re \langle \mathbf{z}, \mathbf{w} \rangle \end{aligned}$$

Second, apply the definition of the induced norm, $\|\mathbf{t}\|^2 := \langle \mathbf{t}, \mathbf{t} \rangle$ and $\|\mathbf{t}\| \geq 0$, to $\mathbf{t} := \mathbf{z} \pm i \cdot \mathbf{w}$:

$$\begin{aligned} \|\mathbf{z} + i \cdot \mathbf{w}\|^2 &= \langle \mathbf{z} + i \cdot \mathbf{w}, \mathbf{z} + i \cdot \mathbf{w} \rangle \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle i \cdot \mathbf{w}, i \cdot \mathbf{w} \rangle + \langle \mathbf{z}, i \cdot \mathbf{w} \rangle + \langle i \cdot \mathbf{w}, \mathbf{z} \rangle \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle i \cdot \mathbf{w}, i \cdot \mathbf{w} \rangle + i \cdot \langle \mathbf{z}, \mathbf{w} \rangle + \overline{i \cdot \langle \mathbf{z}, \mathbf{w} \rangle} \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle i \cdot \mathbf{w}, i \cdot \mathbf{w} \rangle + 2 \cdot i \cdot i \cdot \Im \langle \mathbf{z}, \mathbf{w} \rangle \\ \|\mathbf{z} - i \cdot \mathbf{w}\|^2 &= \langle \mathbf{z} - i \cdot \mathbf{w}, \mathbf{z} - i \cdot \mathbf{w} \rangle \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle i \cdot \mathbf{w}, i \cdot \mathbf{w} \rangle - \langle \mathbf{z}, i \cdot \mathbf{w} \rangle - \langle i \cdot \mathbf{w}, \mathbf{z} \rangle \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle i \cdot \mathbf{w}, i \cdot \mathbf{w} \rangle - i \cdot \langle \mathbf{z}, \mathbf{w} \rangle - \overline{i \cdot \langle \mathbf{z}, \mathbf{w} \rangle} \\ &= \langle \mathbf{z}, \mathbf{z} \rangle + \langle i \cdot \mathbf{w}, i \cdot \mathbf{w} \rangle - 2 \cdot i \cdot i \cdot \Im \langle \mathbf{z}, \mathbf{w} \rangle \\ \|\mathbf{z} + i \cdot \mathbf{w}\|^2 - \|\mathbf{z} - i \cdot \mathbf{w}\|^2 &= 4 \cdot i \cdot i \cdot \Im \langle \mathbf{z}, \mathbf{w} \rangle = -4 \cdot \Im \langle \mathbf{z}, \mathbf{w} \rangle \end{aligned}$$

Example 3 The complex conjugate function $g : \mathbb{C} \rightarrow \mathbb{C}, z \mapsto \bar{z}$ preserves magnitudes but is *anti-linear*: $|g(z)| = |\bar{z}| = |z|$, but $g(i \cdot z) = \overline{i \cdot z} = \bar{i} \cdot \bar{z} = -i \cdot g(z) = -i \cdot g(z) \neq i \cdot g(z)$ for every $z \neq 0$.

Also, $|\langle g(z), g(w) \rangle| = |\langle \bar{z}, \bar{w} \rangle| = |\bar{z} \cdot \bar{w}| = |\overline{z \cdot w}| = |\overline{z \cdot w}| = |\bar{z} \cdot w| = |\langle z, w \rangle|$.

Theorem 4 If $G : \mathbb{C}^n \rightarrow \mathbb{C}^n$ and $\|G(\mathbf{z}) - G(\mathbf{w})\| = \|\mathbf{z} - \mathbf{w}\|$ with $G(i \cdot \mathbf{w}) = i \cdot G(\mathbf{w})$ for all $\mathbf{z}, \mathbf{w} \in \mathbb{C}^n$, then G is linear.

Proof. This proof is adapted for complex linear spaces from [3, §4.2, p. 125–126].

Setting $\mathbf{w} := \mathbf{0}$ gives $G(\mathbf{0}) = G(i \cdot \mathbf{0}) = i \cdot G(\mathbf{0})$, whence $(1 - i) \cdot G(\mathbf{0}) = \mathbf{0}$ and $G(\mathbf{0}) = \mathbf{0}$. Hence, $\|G(\mathbf{z})\| = \|G(\mathbf{z}) - G(\mathbf{0})\| = \|\mathbf{z} - \mathbf{0}\| = \|\mathbf{z}\|$ for every $\mathbf{z} \in \mathbb{C}^n$. Consequently,

$$\begin{aligned} 2 \cdot \Re \langle G(\mathbf{z}), G(\mathbf{w}) \rangle &= \|G(\mathbf{z})\|^2 + \|G(\mathbf{w})\|^2 - \|G(\mathbf{z}) - G(\mathbf{w})\|^2 \\ &= \|\mathbf{z}\|^2 + \|\mathbf{w}\|^2 - \|\mathbf{z} - \mathbf{w}\|^2 = 2 \cdot \Re \langle \mathbf{z}, \mathbf{w} \rangle \end{aligned}$$

Also, $\|G(c \cdot \mathbf{w})\| = \|c \cdot \mathbf{w}\| = |c| \cdot \|\mathbf{w}\| = |c| \cdot \|G(\mathbf{w})\| = \|c \cdot G(\mathbf{w})\|$ for all $c \in \mathbb{C}$ and $\mathbf{w} \in \mathbb{C}^n$. If G is a transformation such that $G(i \cdot \mathbf{w}) = i \cdot G(\mathbf{w})$ for every $\mathbf{w} \in \mathbb{C}^n$, then

$$\begin{aligned} -2 \cdot \Im \langle G(\mathbf{z}), G(\mathbf{w}) \rangle &= \|G(\mathbf{z})\|^2 + \|i \cdot G(\mathbf{w})\|^2 - \|G(\mathbf{z}) - i \cdot G(\mathbf{w})\|^2 \\ &= \|G(\mathbf{z})\|^2 + \|G(i \cdot \mathbf{w})\|^2 - \|G(\mathbf{z}) - G(i \cdot \mathbf{w})\|^2 \\ &= \|\mathbf{z}\|^2 + \|i \cdot \mathbf{w}\|^2 - \|\mathbf{z} - i \cdot \mathbf{w}\|^2 = -2 \cdot \Im \langle \mathbf{z}, \mathbf{w} \rangle \end{aligned}$$

Therefore, $\langle G(\mathbf{z}), G(\mathbf{w}) \rangle = \langle \mathbf{z}, \mathbf{w} \rangle$ for all $\mathbf{z}, \mathbf{w} \in \mathbb{C}^n$. In particular, if $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ is an ordered orthonormal basis of $\mathbb{C}^n$, and $\mathbf{w}_k := G(\mathbf{e}_k)$ for every k , then $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ is also an ordered orthonormal basis of $\mathbb{C}^n$, because $\langle \mathbf{w}_k, \mathbf{w}_\ell \rangle = \langle G(\mathbf{e}_k), G(\mathbf{e}_\ell) \rangle = \langle \mathbf{e}_k, \mathbf{e}_\ell \rangle$ for all k and ℓ . Consequently,

$$\begin{aligned} \langle \mathbf{w}_k, G(c \cdot \mathbf{z}) \rangle &= \langle G(\mathbf{e}_k), G(c \cdot \mathbf{z}) \rangle = \langle \mathbf{e}_k, c \cdot \mathbf{z} \rangle = c \cdot \langle \mathbf{e}_k, \mathbf{z} \rangle \\ &= c \cdot \langle G(\mathbf{e}_k), G(\mathbf{z}) \rangle \\ &= c \cdot \langle \mathbf{w}_k, G(\mathbf{z}) \rangle \end{aligned}$$

for every k , whence $G(c \cdot \mathbf{z}) = c \cdot G(\mathbf{z})$. Similarly,

$$\begin{aligned} \langle \mathbf{w}_k, G(\mathbf{z} + \mathbf{t}) \rangle &= \langle G(\mathbf{e}_k), G(\mathbf{z} + \mathbf{t}) \rangle = \langle \mathbf{e}_k, \mathbf{z} + \mathbf{t} \rangle = \langle \mathbf{e}_k, \mathbf{z} \rangle + \langle \mathbf{e}_k, \mathbf{t} \rangle \\ &= \langle G(\mathbf{e}_k), G(\mathbf{z}) \rangle + \langle G(\mathbf{e}_k), G(\mathbf{t}) \rangle \\ &= \langle \mathbf{w}_k, G(\mathbf{z}) \rangle + \langle \mathbf{w}_k, G(\mathbf{t}) \rangle \end{aligned}$$

for every k , whence $G(\mathbf{z} + \mathbf{t}) = G(\mathbf{z}) + G(\mathbf{t})$. Therefore, G is a linear transformation.

Theorem 5 If $\|U \cdot \mathbf{w}\| = \|\mathbf{w}\|$ for every $\mathbf{w} \in \mathbb{C}^n$, then $\langle U \cdot \mathbf{z}, U \cdot \mathbf{w} \rangle = \langle \mathbf{z}, \mathbf{w} \rangle$ for all $\mathbf{z}, \mathbf{w} \in \mathbb{C}^n$.

Proof. If $G(\mathbf{w}) = U \cdot \mathbf{w}$, then $G(i \cdot \mathbf{w}) = U \cdot i \cdot \mathbf{w} = i \cdot U \cdot \mathbf{w} = i \cdot G(i \cdot \mathbf{w})$. Use Theorem 4.

Theorem 6 If $\langle U \cdot \mathbf{z}, U \cdot \mathbf{w} \rangle = \langle \mathbf{z}, \mathbf{w} \rangle$ for all $\mathbf{z}, \mathbf{w} \in \mathbb{C}^2$, then $U^* \cdot U = I$.

Proof.

$$\begin{aligned}
 U &= \begin{pmatrix} U_{1,1} & U_{1,2} \\ U_{2,1} & U_{2,2} \end{pmatrix}, \\
 U^* \cdot U &= \begin{pmatrix} U_{1,1}^* & U_{2,1}^* \\ U_{1,2}^* & U_{2,2}^* \end{pmatrix} \cdot \begin{pmatrix} U_{1,1} & U_{1,2} \\ U_{2,1} & U_{2,2} \end{pmatrix} \\
 &= \begin{pmatrix} |U_{1,1}|^2 + |U_{2,1}|^2 & U_{1,1}^* \cdot U_{1,2} + U_{2,1}^* \cdot U_{2,2} \\ U_{1,2}^* \cdot U_{1,1} + U_{2,2}^* \cdot U_{2,1} & |U_{1,2}|^2 + |U_{2,2}|^2 \end{pmatrix}
 \end{aligned}$$

$$0 = \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = \begin{pmatrix} 0 & 1 \end{pmatrix} \cdot U^* \cdot U \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (U^* \cdot U)_{2,1} = U_{1,2}^* \cdot U_{1,1} + U_{2,2}^* \cdot U_{2,1},$$

$$0 = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \cdot U^* \cdot U \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (U^* \cdot U)_{1,2} = U_{1,1}^* \cdot U_{1,2} + U_{2,1}^* \cdot U_{2,2},$$

Thus, the columns of U are mutually perpendicular.

$$1 = \left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\|^2 = \begin{pmatrix} 1 & 0 \end{pmatrix} \cdot U^* \cdot U \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (U^* \cdot U)_{1,1} = |U_{1,1}|^2 + |U_{2,1}|^2,$$

$$1 = \left\| \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\|^2 = \begin{pmatrix} 0 & 1 \end{pmatrix} \cdot U^* \cdot U \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (U^* \cdot U)_{2,2} = |U_{1,2}|^2 + |U_{2,2}|^2$$

Thus, the columns of U are mutually orthonormal: $(U^* \cdot U)_{k,\ell} = I_{k,\ell}$, whence $U^* \cdot U = I$.

Since $|U_{1,1}|^2 + |U_{2,1}|^2 = 1$, the polar coordinates of the Cartesian point $(|U_{1,1}|, |U_{2,1}|)$ are $(1, \theta)$ with $0 \leq \theta \leq \pi/2$. Thus, $|U_{1,1}| = \cos(\theta)$, $|U_{2,1}| = \sin(\theta)$: $U_{1,1} = e^{i\alpha} \cdot \cos(\theta)$, $U_{2,1} = e^{i\beta} \cdot \sin(\theta)$.

Since $|U_{1,2}|^2 + |U_{2,2}|^2 = 1$, the polar coordinates of the Cartesian point $(|U_{1,2}|, |U_{2,2}|)$ are $(1, \gamma)$ with $0 \leq \gamma \leq \pi/2$. Thus, $|U_{1,2}| = \cos(\gamma)$, $|U_{2,2}| = \sin(\gamma)$: $U_{1,2} = e^{i\delta} \cdot \cos(\gamma)$, $U_{2,2} = e^{i\lambda} \cdot \sin(\gamma)$. Multiplying a column by any $e^{i\phi}$ keeps them orthonormal. Multiply the first column by $e^{-i\alpha}$ and set $\beta' := \beta - \alpha$. Multiply the second column by $e^{-i\lambda}$ and set $\delta' := \delta - \lambda$.

$$\begin{aligned}
 U_{1,1} &= \cos(\theta), & U_{1,2} &= e^{i\delta'} \cdot \cos(\gamma), \\
 U_{2,1} &= e^{i\beta'} \cdot \sin(\theta), & U_{2,2} &= \sin(\gamma).
 \end{aligned}$$

The columns are mutually orthogonal if and only if

$$0 = \cos(\theta) \cdot e^{i\delta'} \cdot \cos(\gamma) + e^{-i\beta'} \cdot \sin(\theta) \cdot \sin(\gamma)$$

Both summands must have the same magnitude in opposite directions: Hence $-\beta' \equiv \delta'$ modulo π , thence $\sin(\gamma) = \pm \cos(\theta)$ and $\cos(\gamma) = \mp \sin(\theta)$:

$$\begin{aligned}
 U_{1,1} &= \cos(\theta), & U_{1,2} &= -e^{i\delta'} \cdot \sin(\theta); & U_{1,1} &= \cos(\theta), & U_{1,2} &= e^{i\delta'} \cdot \sin(\theta), \\
 U_{2,1} &= e^{-i\delta'} \cdot \sin(\theta), & U_{2,2} &= \cos(\theta); & U_{2,1} &= e^{-i\delta'} \cdot \sin(\theta), & U_{2,2} &= -\cos(\theta); \\
 1 &= \det(U); & & & -1 &= \det(U), & U^* &= U, 0 = \text{trace}(U).
 \end{aligned}$$

Definition 7 A matrix $U \in \mathbb{M}_{k \times k}(\mathbb{C})$ is *unitary* if and only if $U^* \cdot U = I$. The *special unitary group*, denoted by $SU(k)$, is the set of all unitary matrices $U \in \mathbb{M}_{k \times k}(\mathbb{C})$ such that $\det(U) = 1$.

3 The Waiting Apparatus and the Propagating Operator

A waiting apparatus is merely an apparatus where particles stay while we wait [2, p. 8-7]. For a system in condition $|\psi(t)\rangle$ at time t , denote by $U(t + \Delta t, t) |\psi(t)\rangle := |\psi(t + \Delta t)\rangle$ the condition of the system at time $t + \Delta t$. Since nothing acts on the system, probabilities, equivalently, magnitudes and hence inner products do not change. Therefore, $U(t + \Delta t, t)$ is an isometry, equivalently, a linear unitary operation: $U^*(t + \Delta t, t) \cdot U(t + \Delta t, t) = I$ at all times t and $t + \Delta t$.

If $\Delta t = 0$, then $U = I$, because $U(t, t) |\psi(t)\rangle = |\psi(t)\rangle$. Hence, if U is continuous, then

$$\begin{aligned} 0 = I' &= [U^* \cdot U]' = (U^*)' \cdot U + U^* \cdot U' = (U')^* \cdot U + U^* \cdot U' = (U')^* \cdot I + I^* \cdot U' = (U')^* + U' \\ 0 &= i \cdot [(U')^* + U'] = [(-i \cdot U')^* + i \cdot U'] \end{aligned}$$

Thus, $i \cdot U' = (i \cdot U')^*$, whence $U' = -i \cdot i \cdot U' = -i \cdot (i \cdot U')^*$, where $K := i \cdot U'$ is self adjoint, $K = K^*$. Define $H := \frac{1}{\hbar} \cdot K = i \cdot \frac{1}{\hbar} \cdot U'$. Hence,

$$\begin{aligned} \frac{d}{dt} |\psi(t)\rangle &= \lim_{\Delta t \rightarrow 0} \frac{|\psi(t + \Delta t)\rangle - |\psi(t)\rangle}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{U(t + \Delta t, t) |\psi(t)\rangle - U(t, t) |\psi(t)\rangle}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{U(t + \Delta t, t) - U(t, t)}{\Delta t} |\psi(t)\rangle = U'(t, t) |\psi(t)\rangle = -i \cdot K |\psi(t)\rangle = -i \cdot \frac{1}{\hbar} \cdot H |\psi(t)\rangle \end{aligned}$$

where $H := \frac{1}{\hbar} \cdot K = i \cdot \frac{1}{\hbar} \cdot U'$ is the energy, determined experimentally by physicists [2, Ch. 8].

4 Schrödinger's Equation [2, Ch. 11].

$$i \cdot \hbar \cdot \dot{\psi} = H \cdot \psi$$

5 Pauli Matrices

Definition 8 The Identity matrix and the Pauli Matrices, X, Y, Z , are

$$I := \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X := \sigma_x := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y := \sigma_y := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z := \sigma_z := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

A matrix H is **Hermitian** if and only if $H = H^* := \overline{H}^T$, the transposed complex conjugate.

A matrix U is **unitary** if and only if $U^* \cdot U = I$.

Theorem 9 Every complex matrix is a linear combination, with complex coefficients, of Pauli matrices.

Every Hermitian matrix is a linear combination, with real coefficients, of Pauli matrices.

Every Hermitian matrix with null trace is a linear combination, with real coefficients, of Pauli matrices $\sigma_x, \sigma_y, \sigma_z$.

Every Hermitian matrix with null trace and negative unit determinant is a linear combination, with a unit vector of real coefficients, of Pauli matrices $\sigma_x, \sigma_y, \sigma_z$.

Proof. Theorem 6 yields the converse.

$$\begin{aligned} U &= x \cdot X + y \cdot Y + z \cdot Z \\ &= x \cdot \sigma_x + y \cdot \sigma_y + z \cdot \sigma_z \\ &= x \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + y \cdot \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + z \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} z & x - i \cdot y \\ x + i \cdot y & -z \end{pmatrix} \\ -\det(U) &= z \cdot z + (x + i \cdot y) \cdot (x - i \cdot y) = x^2 + y^2 + z^2 = 1. \end{aligned}$$

Theorem 10 Recall that in $\mathbb{H}_1$ the coordinates of $(|+z\rangle, |-z\rangle) = (\psi_0, \psi_1)$ are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

The coordinates of $(|+x\rangle, |-x\rangle)$ form an ordered basis of eigenvectors for the Pauli matrix X .

The coordinates of $(|+y\rangle, |-y\rangle)$ form an ordered basis of eigenvectors for the Pauli matrix Y .

The coordinates of $(|+z\rangle, |-z\rangle)$ form an ordered basis of eigenvectors for the Pauli matrix Z .

The eigenvalues are 1 and -1 , respectively.

Proof.

$$\begin{aligned} X \cdot |+x\rangle &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}, & X \cdot |-x\rangle &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (-1) \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}; \\ Y \cdot |+y\rangle &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ i \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ i \end{pmatrix}, & Y \cdot |-y\rangle &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -i \end{pmatrix} = (-1) \cdot \begin{pmatrix} 1 \\ -i \end{pmatrix}; \\ Z \cdot |+z\rangle &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}, & Z \cdot |-z\rangle &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (-1) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \end{aligned}$$

The matrices $S_x := \frac{1}{2} \cdot \sigma_x$, $S_y := \frac{1}{2} \cdot \sigma_y$, $S_z := \frac{1}{2} \cdot \sigma_z$, have eigenvalues $\frac{1}{2}$ and $-\frac{1}{2}$.

References

- [1] Johannes A. Buchmann. *Introduction to quantum algorithms*, volume 64 of *Pure and Applied Undergraduate Texts*. American Mathematical Society, Providence, RI, 2024.
- [2] Richard Phillips Feynman, Robert B. Leighton, and Matthew L. Sands. *The Feynman Lectures on Physics*, volume III. Addison-Wesley, Redwood City, CA, New Millennium, Printing 11 edition, 2020. Free versions may be available online.
- [3] Wendell Helms Fleming. *Functions of Several Variables*. Springer-Verlag, New York, NY, second edition, 1987. Corrected third printing; ISBN 0-387-90206-6; QA331.F63 1977; LCCC No. 76-40029; 515'.84.
- [4] Lin Lin and Jianfeng Lu. *A mathematical introduction to electronic structure theory*, volume 4 of *SIAM Spotlights*. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2019.